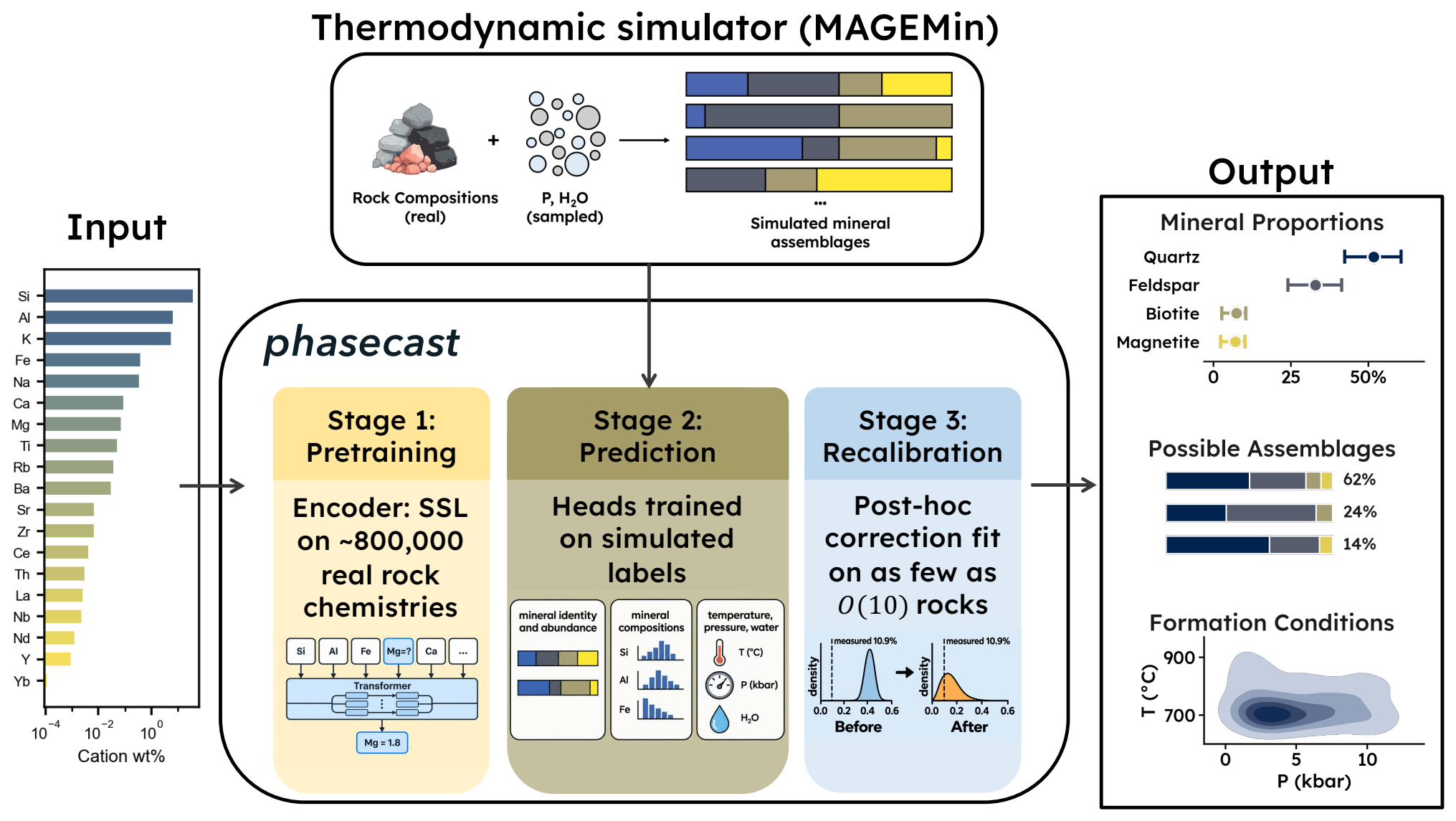

Thermodynamic simulator (MAGEMin)
Rock Compositions (real)
+
P, H2O (sampled)
...
Simulated mineral assemblages
Input
Si
Al
K
Fe
Na
Ca
Mg
Ti
Rb
Ba
Sr
Zr
Ce
Th
La
Nb
Nd
Y
Yb
10−4
10−2
100
Cation wt%
phasecast
Stage 1: Pretraining
Encoder: SSL on ~800,000 real rock chemistries
Si
Al
Fe
Mg=?
Ca
...
Transformer
Mg = 1.8
Stage 2: Prediction
Heads trained on simulated labels
mineral identity and abundance
mineral compositions
Si
Al
Fe
temperature, pressure, water
T (°C)
P (kbar)
H2O
Stage 3: Recalibration
Post-hoc correction fit on as few as $O(10)$ rocks
measured 10.9%
density
0.0 0.2 0.4 0.6
Before
measured 10.9%
density
0.0 0.2 0.4 0.6
After
Output
Mineral Proportions
Quartz
Feldspar
Biotite
Magnetite
0
25
50%
Possible Assemblages
62%
24%
14%
Formation Conditions
T (°C)
900
700
0
5
10
P (kbar)